

EVEN MID SEMESTER– 2025

Name of the Course: B.Tech
EE

Semester: 8th

Name of the Paper: Wind and
Solar Energy Systems

Paper Code: TEE802

Time: 1.5 Hours

Maximum Marks: 50

Note:-

- (i) All questions are compulsory.
- (ii) Answer any one sub questions among a, b in each main question
- (iii) Total marks in each main question are 10.
- (iv) Each question carries 10 marks

Q1	(10marks)	CO1, CO2
(a)	Explain the working principle of a hybrid wind-solar energy system.	
(b)	Discuss the advantages and challenges of integrating wind and solar energy into the power grid.	
Q2	(10 marks)	CO3, CO2
(a)	Describe the role of power electronics in hybrid wind-solar energy systems.	
(b)	Compare and contrast the efficiency and reliability of wind energy and solar energy in different climatic conditions.	
Q3	(10 marks)	CO3
(a)	What are the key components of a wind energy system, and how do it function to generate electricity?	
(b)	Explain the impact of seasonal variations on the performance of a solar system.	
Q4	(10 marks)	CO1
(a)	Explain briefly about components of tidal power plants.	
(b)	The efficiency of a solar cell is 20% and its surface area is 4cm ² . Find the electrical energy generated in one second, by 3000 cells, connected in a solar panel, if 640J of solar energy is incident on one meter square area, in one second.	
Q5	(10 marks)	CO3
(a)	Differentiate between Non-conventional energy sources and conventional energy sources.	
(b)	Analyze the economic feasibility of a wind-solar hybrid system compared to conventional energy sources.	

